Bill St. Clair
Senior Software Engineer
Location: Bennington, Vermont

Seeking

Software design and development using my senior developer skills with opportunities for learning. Telecommuting from home greatly preferred. Erlang/Elixir, Haskell, Elm, & Clojure are my latest studies, but interesting projects are more important than programming language.

Summary

A seasoned programming professional with expertise in privacy protection, encryption, trusted systems, data structures and algorithms. Extensive experience programming in Common Lisp as well as Swift, C, JavaScript, Java, PHP, XML, and HTML. Experience programming Macintosh and iOS with the Cocoa application framework. This resume is on-line at lisplog.org/resume.html.

Selected Professional Experience

Gib Goy Games (2017)

- This is my own brand for my own games: GibGoyGames.com. It grew out of my new-found love of the Elm programming language.
  - Kakuro Dojo is an implementation of the Kakuro numeric crossword puzzle. It runs in web browsers and as an app on phones and tablets, available through the Apple iOS and Google Play app stores. Code at github.com/billstclair/kakuro-master.

Clozure Associates (2006-Present)

- Senior Common Lisp engineer on a variety of large mission-critical projects, including the following:
  - Opusmodus, a Macintosh music composition application (opusmodus.com).
  - Antenna control for a solid-state satellite terminal
  - A multi-user, replicating, memory-resident database.
  - A real-time system for aggregating and canonicalizing data streams.
  - Language translation support, database backend, and web front-end for a disease research web site.
  - Contributed to several functional areas of a large-scale airline reservation system.
- Contributed to iOS and Android application development projects.
  - Steady Type, a custom iOS keyboard, written in Swift.

Shaker Computer and Management Services, Inc. (1997-2006)

- COINS: COnstruction Industry Software, RIS: Residential Information System - designed and wrote assorted code for these Enterprise systems; wrote Technical Manual for RIS.
- A Java front-end for COINS job status information - designed and coded entire system.
- Titanium User Interface Manager (UIM), a generic user interface written in Java using XML to encode screen layouts and InfoBus to pass data around. Java client, Progress 4GL server.

- Wrote and presented a series of Java programming courses to Shaker's programming staff.

Digitool, Inc. (1995-1997)

- Shipped Macintosh Common Lisp 3.9, 4.0, 4.1
- Implemented the shared library interface using CFM
- Ported the inspector, stack backtrace, and persistent object store to new architecture
- Reimplemented stack groups and implemented segmented stacks
- Made CL-HTTP OpenTransport native
- Added primary slot optimizations for CLOS.

Apple Computer, Inc (1989-1995)

- Implemented compiler database persistence, the module system, and the dependency system for the Apple Dylan programming language.
- Implemented the Common Lisp Object System (CLOS) for Macintosh Common Lisp
- Converted the MCL GUI from Object Lisp to CLOS
- Re-implemented hash tables
- Designed and implemented WOOD, a simple persistent object store using Btrees

Thinking Machines Corporation (1984-1986)

- CDL (a circuit design language used to design the CM-2), sticks editor for VLSI design. All on the Lisp Machine.

Personal Projects

- 2016 Kakuro in Elm (kakuro-dojo.com)
- 2016 Conway's Life in JavaScript (billstclair.com/life)
- 2016 Google Authenticator in JavaScript (billstclair.com/two-factor-authenticator)
- 2012 jsMaze, a multi-player maze game in JavaScript, with node.js server (jsmaze.com)
- 2011 Lisplog, a templating system using Hunchentoot to maintain a blog (lisplog.org)
- 2009 Truledger, a digitally-signed general ledger and trading system (truledger.com)
- 2008 loomclient, a PHP client to loom.cc (github.com/billstclair/loomclient)
- 1988 HyperArrays, a multi-dimensional array mechanism for HyperCard.
- 1985 Mandelbrot explorer for Lisp machine.
- 1979 Scheme interpreter for the PDP-11.

Education

- MIT, 1978, BS in Mathematics. GPA 4.9/5.0. Phi Beta Kappa, Sigma Xi, Tau Beta Pi.

Technical Skills

- Proficient in: Common Lisp, Swift, Java, EMACS, Apache
- Practical experience with: Elm, C, JavaScript/Node.js, PHP, Python, Ruby, XML, Unix/Linux, HTML, Mac OS X, iOS, Objective C
- Learning: Haskell, Erlang, Elixir, Phoenix, Riak, Rebar
- Historic proficiencies: Progress 4GL/Database/AppServer, PowerPC assembler, 68K assembler, Macintosh toolbox, Symbolics Lisp Machine, RATFOR, PDP-11 assembler, RT-11, RSX-11M, oscilloscopes, logic analyzers, 8X-300, 2901, TMS-320.
- Historic practical experience: Dylan, Pascal, APL, PL/1, MS-DOS, Multics, FoxBase, RBase